

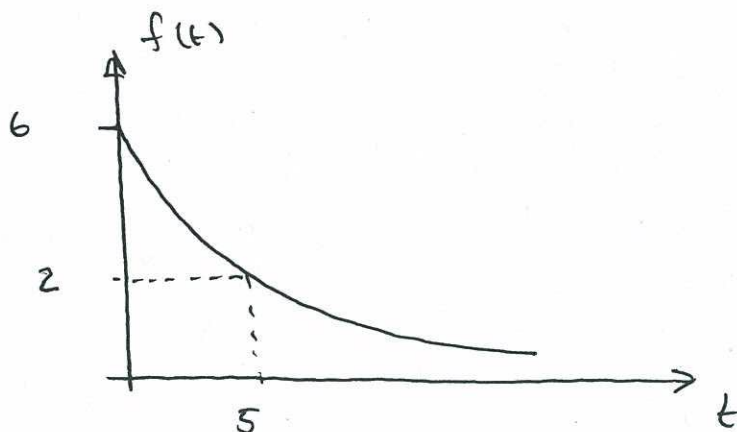
Exercises
chapter 6.1

1. The steady-state response is the response as $t \rightarrow \infty$. In this case, the velocity of the mass as $t \rightarrow \infty$ is zero (it comes to rest).

$$\underline{\underline{v(t \rightarrow \infty) = 0 \text{ m/s}}}$$

1. $f(t) = u(t+1) - 3u(t-1) + 2u(t-3)$

2.



Form of exponential: $f(t) = A_0 e^{-t/\tau}$

From graph, $A_0 \approx 6$
 $\tau \approx 5 \text{ msec}$ } $\Rightarrow f(t) \approx 6 e^{-t/5 \times 10^{-3}}$

$f(t) \approx 6 e^{-200t}$

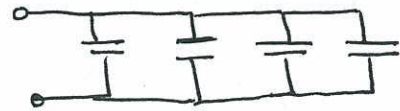
Exercises
Chapter 6.3

1.

- maximum capacitance from parallel combination (capacitances add directly)

$$C_{\max} = 4 \mu\text{F}$$

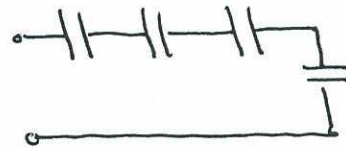
$\Rightarrow$



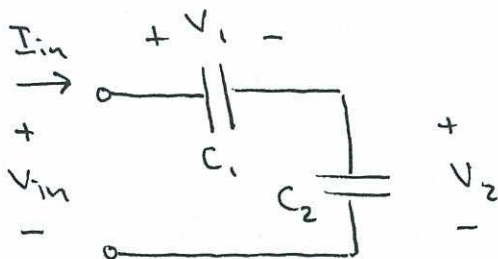
- minimum capacitance from series combination:

$$C_{\text{eq}} = 0.25 \mu\text{F}$$

$\Rightarrow$



2.



$$I_{\text{in}} = \frac{C_1 C_2}{C_1 + C_2} \frac{dV_{\text{in}}}{dt}$$

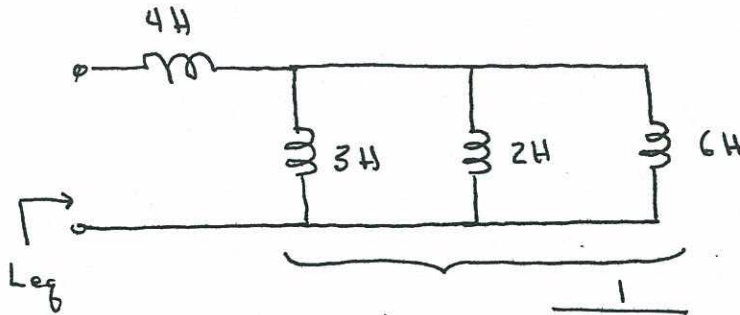
$$V_1 = \frac{1}{C_1} \int I_{\text{in}} dt + V_{\text{in}}(0) \Rightarrow \underline{\underline{V_1 = \frac{C_2}{C_1 + C_2} V_{\text{in}}}}$$

Likewise, $\underline{\underline{V_2 = \frac{C_1}{C_1 + C_2} V_{\text{in}}}}$

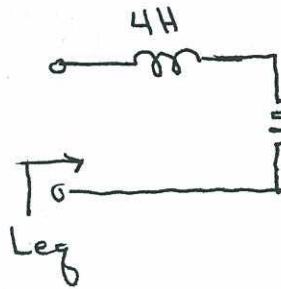
If $C_1 = C_2 = 10 \mu\text{F}$, $\underline{\underline{V_2 = \frac{1}{2} V_{\text{in}}}}$

Exercises
Chapter 6.4

1.



$$L_{eq_1} = \frac{1}{\frac{1}{3H} + \frac{1}{2H} + \frac{1}{6H}} = 4H$$



$$\Rightarrow L_{eq} = 4H + 4H = \underline{\underline{8H}}$$